Experiment 14: Pepsi Black — CO_2 SVEs Mimicking Dolphins in the Black Sea

"Gas Bubbles Mimicking Dolphins in the Black Sea"

Experimental Protocol

Container: Standard glass cup (copo de vidro normal). **Liquid:** Pepsi Black, chilled (gelada), freshly served. **Heat source:** None — ambient conditions only. **Recording:** iPhone slow-motion video, frame-by-frame analysis. **Analysis:** Single SVE tracked across 40 consecutive frames (5 collages × 8 frames) from pre-emergence through aerial arc to re-entry into the liquid. SVE position annotated with red circle in each frame. **Video:** https://youtu.be/FBsfnDhfJIk

Complete Data: Experiment 14 (5 Collages, 40 Frames)

Coll.	Fr	Frame #	Time	Phase	Observation
C1	F1	226	07:915	CONTEXT	Context frame. SVE (red circle) visible deep inside the liquid, well below the surface. Numerous other bubbles/SVEs visible on glass walls. This frame establishes the SVE's starting position — submerged.
C1	F2	433	14:400	SUBMERGED	SVE (red circle) now near the liquid surface, approaching from below. Ascending trajectory visible. Multiple other SVEs adhered to glass walls.
C1	F3	434	14:433	SUBMERGED	SVE continues upward approach toward the liquid-air interface.
C1	F4	435	14:467	INTERFACE	SVE reaches the liquid surface. Contact with liquid-air interface. Classical prediction: bubble should rupture here (surface tension breach).
C1	F5	436	14:500	EMERGING	SVE breaches the surface. Maintains spherical integrity — does NOT rupture upon crossing the interface.
C1	F6	437	14:533	EMERGING	SVE continues upward past the surface. Clearly above the liquid level. Intact.
C1	F7	438	14:567	AIRBORNE	SVE now airborne above the liquid surface. Maintains cohesion — solid/semi-solid behavior.
C1	F8	439	14:600	AIRBORNE	SVE continues ascending. Red circle tracks upward displacement. Trajectory: rising arc, moving from left toward center of frame.
C2	F9	440	14:633	AIRBORNE	SVE ascending — now at significant height above the liquid surface. Arc trajectory continues upward and rightward.
C2	F10	441	14:667	AIRBORNE	SVE near peak height. Trajectory flattening — approaching apex of arc.
C2	F11	442	14:700	APEX	SVE at or near maximum height. Red circle shows position well above the liquid surface — highest point in the dolphin-like arc.
C2	F12	443	14:733	APEX/DESC	SVE begins descent from apex. Trajectory curving downward and rightward. Maintains structural integrity throughout.
C2	F13	444	14:767	DESCENDING	SVE descending. Arc trajectory now clearly curving back toward the liquid.
C2	F14	445	14:833	DESCENDING	SVE continues downward arc. Approaching the liquid surface from above.
C2	F15	446	14:867	DESCENDING	SVE lower — nearing the liquid-air interface on descent.
C2	F16	447	14:900	DESCENDING	SVE approaching re-entry point. Still intact, still spherical.
C3	F17	448	14:933	DESCENDING	SVE continuing descent toward the liquid surface.
C3	F18	449	14:967	DESCENDING	SVE approaching the liquid surface. Red circle tracks downward displacement.
C3	F19	450	15:000	DESCENDING	SVE very close to the liquid surface. Imminent re-entry.

Coll.	Fr	Frame #	Time	Phase	Observation
C3	F20	451	15:033	INTERFACE	SVE contacts the liquid surface from above. Classical prediction: should merge/pop upon contact.
C3	F21	452	15:067	RE-ENTERING	SVE penetrates back INTO the liquid. Maintains spherical form — does NOT burst upon re-entry. The "dive."
C3	F22	453	15:100	RE-ENTERING	SVE submerging — now partially below the surface. Structural integrity fully maintained.
C3	F23	454	15:133	SUBMERGED	SVE now below the surface. Re-entry complete. Red circle shows position inside the liquid.
C3	F24	455	15:167	SUBMERGED	SVE descending deeper into the liquid after re-entry.
C4	F25	456	15:200	SUBMERGED	SVE sinking further inside the liquid.
C4	F26	457	15:233	SUBMERGED	SVE continues downward inside the liquid.
C4	F27	458	15:267	SUBMERGED	SVE deeper. Red circle shows continued descent inside the liquid.
C4	F28	459	15:300	SUBMERGED	SVE descending further. Behavior consistent with a denser-than-liquid entity — if it were a gas bubble, buoyancy would push it back to the surface.
C4	F29	460	15:333	SUBMERGED	SVE deep inside the liquid. Position now lower than initial ascent point.
C4	F30	461	15:367	SUBMERGED	SVE continues sinking. Gravitational behavior — not buoyant.
C4	F31	462	15:400	SUBMERGED	SVE deep. Red circle near bottom of visible liquid.
C4	F32	463	15:433	SUBMERGED	SVE at depth. Maintaining structural integrity throughout post-dive submersion.
C5	F33	464	15:467	SUBMERGED	SVE deep inside the liquid. Fading from trackable visibility.
C5	F34	465	15:500	SUBMERGED	SVE still visible but diminishing in contrast against dark liquid.
C5	F35	466	15:533	SUBMERGED	SVE barely distinguishable. Descending or dissolving at depth.
C5	F36	467	15:567	DEEP	SVE at maximum tracked depth. Red circle annotation still marks position.
C5	F37	469	15:600	DEEP	SVE nearly lost to visibility.
C5	F38	470	15:633	DEEP	SVE at tracking limit.
C5	F39	471	15:667	FADING	SVE disappearing into the dark liquid.
C5	F40	472	15:700	GONE	SVE no longer trackable. Entity has fully re-submerged and disappeared into the Pepsi Black. Complete dolphin cycle: ascent → breach → arc → dive → deep submersion → disappearance.

Photographic Documentation

Figure 56: E14c1 — CONTEXT + PRE-EMERGENCE TO AIRBORNE (8 frames). Frame 226 (context): SVE submerged deep inside the Pepsi Black. Frames 433–435: SVE ascends toward the liquid-air interface. Frame 435: contact with surface — classical prediction: rupture. Frames 436–439: SVE breaches surface, maintains spherical integrity, becomes airborne. NO rupture upon surface tension breach.

Figure 57: E14c2 — MAXIMUM HEIGHT ARC (8 frames). Frames 440–442: SVE ascending through air above the liquid — reaching peak height (frame 442). Frames 443–447: SVE at apex, begins descent. Dolphin-like arc trajectory clearly visible across the 8-frame sequence. Structural integrity maintained throughout airborne phase.

Photographic Documentation (continued)

Figure 58: E14c3 — DESCENT + RE-ENTRY + DIVE (8 frames). Frames 448–450: SVE descending back toward the liquid surface. Frame 451: contact with liquid from above — classical prediction: merger/pop. Frames 452–453: SVE penetrates BACK INTO the liquid — the "dive." Maintains spherical form through re-entry. Frames 454–455: SVE now submerged, descending into the liquid.

Figure 59: E14c4 — POST-DIVE DEEP SUBMERSION (8 frames). Frames 456–463: SVE continues sinking deeper into the Pepsi Black. Gravitational descent — NOT buoyant. If this were a gas bubble (density < liquid), buoyancy would force it back to the surface. Instead, it sinks. Behavior consistent with a solid or semi-solid entity denser than the liquid.

Photographic Documentation (continued)

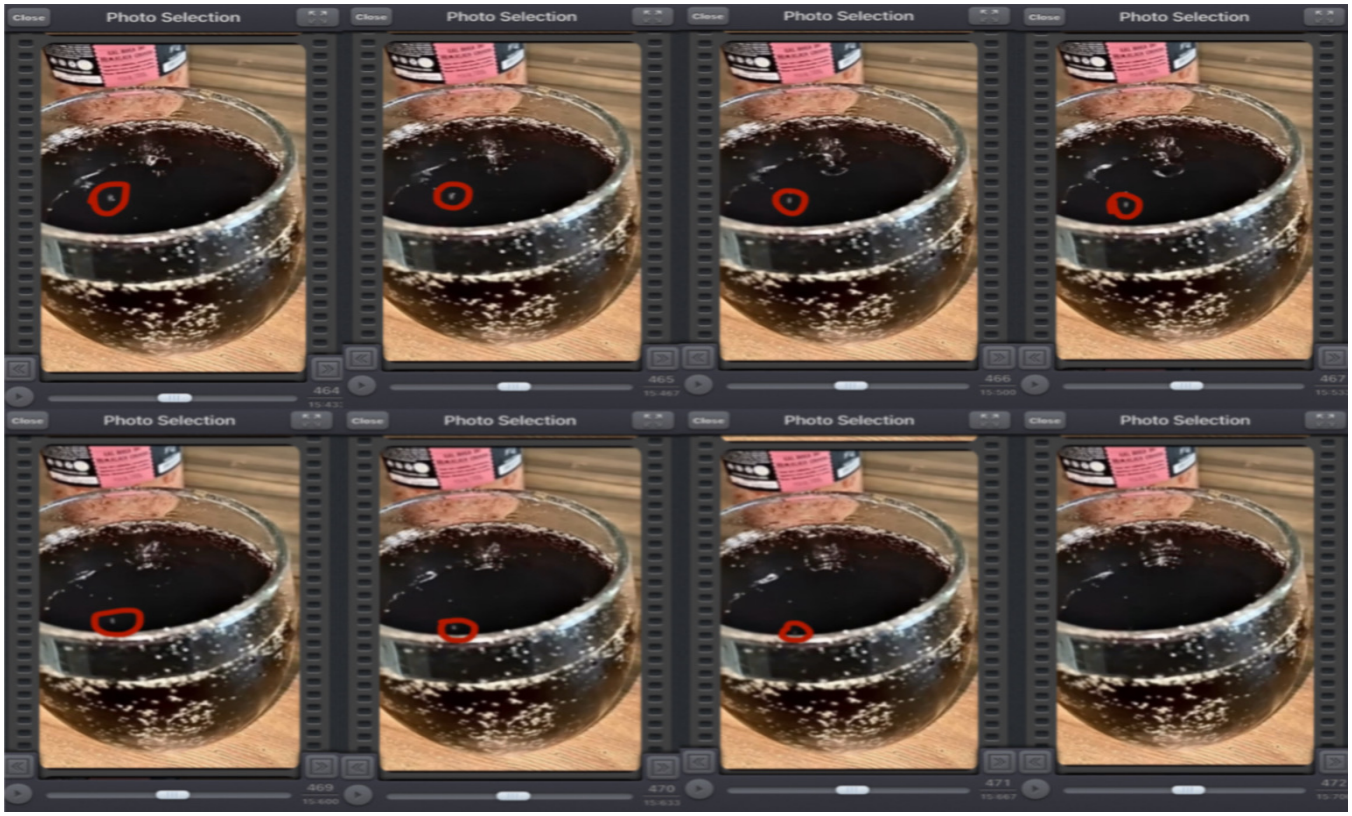

Figure 60: E14c5 — DEEP SUBMERSION + DISAPPEARANCE (8 frames). Frames 464–467: SVE deep inside the dark liquid, fading from visibility. Frames 469–472: SVE disappears entirely — lost in the "Black Sea." Complete dolphin cycle documented: ascent → surface breach → airborne arc → apex → descent → re-entry dive → deep submersion → disappearance. Total span: ~7.8 seconds of slow-motion footage (frames 226–472).

Anomaly Analysis — Experiment 14

Anomaly E1: Surface tension breach without rupture.

At frame 435–436, the SVE crosses the liquid-air interface from below — breaching what classical fluid dynamics defines as the surface tension barrier. For a gas bubble, crossing this interface means the thin liquid film surrounding the gas pocket is stretched beyond its mechanical limit and ruptures, releasing the gas. This is the *raison d'être* of surface tension in the bubble lifecycle: the bubble exists because surface tension traps gas; when the bubble reaches the surface, the film thins and breaks. Here, the entity crosses the interface and maintains its spherical shape entirely intact. This is structurally impossible for a gas bubble — it is consistent only with a solid or semi-solid entity (SVE) whose cohesion is mechanical/structural, not film-dependent.

Anomaly E2: Ballistic arc trajectory — the "dolphin jump."

Across frames 436–455, the SVE executes a complete parabolic arc: ascent → apex → descent → re-entry. This trajectory is ballistic — governed by inertia and gravity, with the entity possessing sufficient momentum at surface breach to carry it through the air and back. A gas bubble, upon rupturing at the surface, disperses its contents into the atmosphere. It does not arc. It does not re-enter. The observed trajectory requires three properties simultaneously: (1) structural integrity to survive the breach, (2) sufficient mass to follow a ballistic parabola, and (3) sufficient cohesion to survive re-entry impact. None of these are properties of free CO_2 gas.

Anomaly E3: Re-entry without merger or dissolution.

At frames 451–453, the SVE impacts the liquid surface from above and penetrates back into the liquid. Classical fluid dynamics predicts two possible outcomes for a gas bubble contacting a liquid surface from above: (a) it sits on the surface supported by surface tension (floating), or (b) it merges with the gas phase above and the liquid closes. Instead, the SVE dives below the surface maintaining structural integrity — identical to a solid object (a stone, a ball) entering water. The "skipping stone" and "diving dolphin" behaviors observed throughout this video are macroscopic signatures of entities with mechanical structure, not gas pockets.

Anomaly E4: Post-dive sinking — density greater than liquid.

After re-entry (frames 456–472), the SVE does not return to the surface. It sinks progressively deeper until it disappears. For a CO_2 gas bubble (density ≈ 1.8 kg/m^3 vs. Pepsi liquid ≈ 1040 kg/m^3), buoyancy would produce an upward force ~577 times the bubble's weight. Sinking is physically impossible for a gas bubble in this system. The observed sinking requires the entity's effective density to exceed that of the surrounding liquid — consistent with a structured vesicular entity (SVE) whose internal composition includes liquid/solid phases, not free gas.

Anomaly E5: Multiple SVEs exhibiting solid-body behaviors simultaneously.

The tracked SVE is not unique — the full video shows dozens of similar entities walking across the surface, bouncing ("skipping stones on a lake"), jumping, and diving. This is not an isolated artifact; it is a population-level phenomenon. The variety of behaviors observed — surface walking, bouncing, jumping, diving — constitutes a behavioral repertoire impossible for gas bubbles and consistent only with solid or semi-solid structures interacting mechanically with the liquid-air interface.

Systematic Falsification — Experiment 14

Classical Prediction	Observation	Status
Gas bubbles rupture upon breaching the liquid-air interface (surface tension film breaks)	SVE crosses the interface intact, maintains spherical form, becomes airborne	**FALSIFIED**
Upon rupture, gas disperses into the atmosphere irreversibly	SVE executes a complete ballistic arc (ascent → apex → descent) and returns to the liquid	**FALSIFIED**

Classical Prediction	Observation	Status
A gas entity contacting liquid from above either floats or merges — it cannot re-enter	SVE dives back into the liquid maintaining structural integrity through re-entry impact	**FALSIFIED**
CO_2 gas density (≈1.8 kg/m^3) << liquid density (≈1040 kg/m^3): bubbles always float	SVE sinks after re-entry — descends progressively deeper until disappearing	**FALSIFIED**
Carbonated beverage "fizz" consists of free CO_2 gas bubbles	Population-level solid-body behaviors: walking, bouncing, jumping, diving — impossible for gas	**FALSIFIED**

Cross-Experiment Consistency

Pattern	Previous Experiments	Experiment 14
SVEs exhibit structural integrity beyond gas-phase predictions	Exp. 9: microscopic SVEs in saliva show thermal attraction, cohesion, motility. Exp. 11–12: vapor anomalies imply structured entities, not free molecules.	SVE survives surface breach, airborne arc, and re-entry impact — mechanical cohesion incompatible with gas phase.
Entities defy buoyancy predictions	Exp. 1: butter climbs inclined planes against gravity. Exp. 7: Leidenfrost orbital trapping.	CO_2 "bubbles" sink in liquid 577× denser — buoyancy force ignored/overridden.
Population-level anomalous behavior	Exp. 9: SVE populations in saliva. Exp. 2: sustained triple-point coexistence far from classical equilibration.	Dozens of SVEs simultaneously walking, bouncing, jumping, diving — population-level solid-body mechanics in a carbonated beverage.
Universality across systems	Water (Exp. 3–4), legumes (Exp. 5–6), herbs (Exp. 11), egg whites (Exp. 13), saliva (Exp. 9), butter (Exp. 1).	Carbonated beverage (Pepsi Black) — seventh distinct system exhibiting SVE-mediated anomalies.

Experiment 14 introduces the most visually dramatic demonstration of SVE structural integrity to date. The complete "dolphin cycle" — a single entity ascending through liquid, breaching the surface intact, executing a ballistic arc through air, re-entering the liquid without rupture, and then sinking to depth — falsifies in a single trajectory four independent classical predictions about gas bubble behavior. The observation requires no thermal gradient, no exotic materials, no specialized equipment: a glass of cold Pepsi and an iPhone in slow-motion mode. The fact that this phenomenon is reproducible with a commercial carbonated beverage purchased at any convenience store makes it perhaps the most accessible and falsifiable experiment in the Kitchen Thermodynamics series. The "CO_2 bubbles" in your soda are not gas bubbles. They are structured vesicular entities — and they behave like dolphins in a black sea.

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Supplementary materials: Google Drive · YouTube: @5incere5cience · TikTok: @5incere5cience
Video: https://youtu.be/FBsfnDhfJIk

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